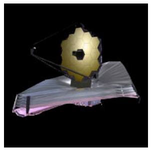

10482 - Crushing It: Tracking the Survival of Cool Clouds on Cloud-Crushing and Dynamical Timescales in the M82 Outflow

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
MSA on Wind				
	8	Leakcal	NIRSpec MultiObject Spectroscopy	(5) M82_MSA_Regions_1
	11	Ref Stars Only v6	NIRSpec MultiObject Spectroscopy	(10) reference_stars_catalog_80
	12	Ref Stars Only v8	NIRSpec MultiObject Spectroscopy	(10) reference_stars_catalog_80

ABSTRACT

The cool components of galaxy winds carry most of the outflowing mass, and are thus key to understanding feedback in the baryon cycle. Despite recent improvements and insights, theory and simulations fail to match observations of cool gas in galaxy outflows, and struggle to explain how cool clouds are entrained and accelerated in the hot flow without being destroyed. JWST imaging of the M82 outflow, one of the closest and most spectacular winds, has revealed its complex filamentary structure. Acquiring the corresponding spectroscopic information, however, is key to inform theory and simulations. This program uses the NIRSpec MSA at high spectral resolution (R~3000) to efficiently measure the kinematics and physical state of cool clouds entrained in M82’s wind at 2.5 pc resolution. We orient the MSA so that the aluminum bar blocks the bright starburst and open shutters across both the northern and southern outflows informed by existing JWST imaging, probing nearly 100 regions up to 2 kpc away from the starburst. The resulting kinematics constrain both dynamical and cloud-crushing timescales all while tracing cloud-destroying shocks, measuring

electron densities and temperatures, and tracing dust processing as a proxy of cool gas destruction. These observations provide a parsec-scale view of cool cloud survival and the evolution of multiple gas phases, which are key inputs for future galactic wind models.

OBSERVING DESCRIPTION

In this program, we will carry out the following observations. The goal is to obtain gas kinematics, cloud-crushing and dynamical timescales, shock diagnostics, and ionized gas temperatures of cool clouds using high-resolution (R=2700) NIRSpec MSA spectroscopy throughout the inner M82 wind.

Observing Strategy: We set up two identical science pointings that only differ in which shutters are open. We position the MSA so that Q1 covers the northern wind and Q2 covers the southern wind. We place the aluminum bar over the bright midplane to minimize the effect of light leakage. This also allows for one fixed slit spectrum "for free" in the M82 midplane. We set up a 3-shutter slitlet configuration for each source, and allow nodding in the slitlets. We select the midpoint constraint and allow for multiple "sources" to fall into slitlet since we are targetting extended filamentary regions. We choose not to dither to prioritize the number of regions of interest, which drastically decreases when introducing only a 2-shutter dither. We constrain the PA angle in order to ensure that the aluminum bar stays fixed over the bright galaxy midplane, and that we capture at least 80% of our targetted regions, with the optimal position falling in the middle of our PA range. For both pointings, we open shutters along bright filaments, with a handful of regions containing little to no emission for proper comparison. We select the G235H/F170LP and G395H/F290LP grating/filter-wheel combinations at the highest resolution (R=2700) to precisely determine velocity centroids and dispersions.

Leakcal: We set up a third pointing, again with the same orientation and parameters as before, but with all shutters set to closed. This will help correct for light leakage from stuck-open shutters.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	M82_MSA_Regions_1	RA: 09 55 19.5112 (148.8312967d) Dec: +69 38 45.04 (69.64584d) Equinox: J2000		
Comments: *Description=[]*				
(6)	M82_MSA_Regions_2	RA: 09 55 45.2583 (148.9385762d) Dec: +69 40 55.76 (69.68216d) Equinox: J2000		
Comments: *Description=[]*				
(9)	reference_stars_catalog_160	RA: 09 55 47.6668 (148.9486117d) Dec: +69 40 32.10 (69.67558d) Equinox: J2000		
Comments: *Description=[]*				
(10)	reference_stars_catalog_80	RA: 09 55 47.6668 (148.9486117d) Dec: +69 40 32.10 (69.67558d) Equinox: J2000		
Comments: *Description=[]*				

Observation

Proposal 10482, Observation 8: Leakcal

Tue May 05 19:00:59 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Comments: While the aperture position angle used for the MSA planning tool is specifically 328.4 degrees, we have tested a range of APAs acceptable for our program. The APA range is in the special requirements (325 - 332 degrees).

Diagnostics

(Leakcal (Obs 8)) Warning (Form): NIRSpec MOS science exposures should not normally use the ALLOPEN or ALLCLOSED MSA configurations.

(Leakcal (Obs 8)) Warning (Form): NIRSpec MOS science exposures should not normally use the ALLOPEN or ALLCLOSED MSA configurations.

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	M82_MSA_Regions_1	RA: 09 55 19.5112 (148.8312967d) Dec: +69 38 45.04 (69.64584d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	M82_MSA_Regions_1 (1470 sources)		jwst-nirspec-hr	1.5

Reference Stars

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (G395H/F290LP)	ALLCLOSED	3 Shutter Slitlet	148.99968554166665 Degrees 69.6876775 Degrees	328.4357701543709			3	3	2609.035
2	2 (G235H/F170LP)	ALLCLOSED	3 Shutter Slitlet	148.99968554166665 Degrees 69.6876775 Degrees	328.4357701543709			3	3	2609.035

Special Requirements	
	Aperture PA Range 325.0 to 332.0 Degrees (V3 186.4254303 to 193.4254303) MSA Planned Aperture PA 328.4000 to 328.4000 Degrees (V3 189.8254303 to 189.8254303)

Observation

Proposal 10482, Observation 11: Ref Stars Only v6

Tue May 05 19:00:59 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(Ref Stars Only v6 (Obs 11)) Warning (Form): Interleaving MSA configurations in a visit increases MSA shutter configuration changes.
(Ref Stars Only v6 (Obs 11)) Warning (Form): NIRSpec MOS science exposures should not normally use the ALLOPEN or ALLCLOSED MSA configurations.
(Ref Stars Only v6 (Obs 11)) Warning (Form): NIRSpec MOS science exposures should not normally use the ALLOPEN or ALLCLOSED MSA configurations.
(Ref Stars Only v6 (Obs 11)) Warning (Form): NIRSpec MOS science exposures should not normally use the ALLOPEN or ALLCLOSED MSA configurations.
(Ref Stars Only v6 (Obs 11)) Warning (Form): NIRSpec MOS science exposures should not normally use the ALLOPEN or ALLCLOSED MSA configurations.
(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(10)	reference_stars_catalog_80	RA: 09 55 47.6668 (148.9486117d) Dec: +69 40 32.10 (69.67558d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	Filter: F110W; Readout: NRSRAPIDD6; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	F110W	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	reference_stars_catalog_80 (160 sources)		jwst-nirspec-hr	1.5

Reference Stars

Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
1	30	148.987703	69.677960	21.504	1	83	148.920333	69.664399	21.506
1	43	148.900809	69.676740	21.506	1	125	148.992193	69.676730	21.502
1	47	148.895547	69.678695	21.508	1	146	148.999060	69.681890	21.503
1	51	148.897363	69.677282	21.514	1	160	148.994476	69.676761	21.504

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (G395H/F290LP)	c1		148.92646404166666 Degrees 69.67284083333334 Degrees	328.37924634585477			1	1	860.745
2	2 (G235H/F170LP)	c1		148.92646404166666 Degrees 69.67284083333334 Degrees	328.37924634585477			1	1	860.745
3	1 (G395H/F290LP)	ALLCLOSED		148.92646404166666 Degrees 69.67284083333334 Degrees	328.37924634585477			1	1	860.745
4	2 (G235H/F170LP)	ALLCLOSED		148.92646404166666 Degrees 69.67284083333334 Degrees	328.37924634585477			1	1	860.745
5	1 (G395H/F290LP)	c2_refstars_shutters		148.92428466666667 Degrees 69.67153944444445 Degrees	328.3772039723182			1	1	860.745
6	2 (G235H/F170LP)	c2_refstars_shutters		148.92428466666667 Degrees 69.67153944444445 Degrees	328.3772039723182			1	1	860.745
7	1 (G395H/F290LP)	ALLCLOSED		148.92428466666667 Degrees 69.67153944444445 Degrees	328.3772039723182			1	1	860.745
8	2 (G235H/F170LP)	ALLCLOSED		148.92428466666667 Degrees 69.67153944444445 Degrees	328.3772039723182			1	1	860.745

Special Requirements

Aperture PA Range 323.5745697 to 345.5745697 Degrees (V3 185.0 to 207.0)
MSA Planned Aperture PA 328.4000 to 328.4000 Degrees (V3 189.8254303 to 189.8254303)

Observation

Proposal 10482, Observation 12: Ref Stars Only v8

Tue May 05 19:00:59 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(10)	reference_stars_catalog_80	RA: 09 55 47.6668 (148.9486117d) Dec: +69 40 32.10 (69.67558d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	Filter: CLEAR; Readout: NRSRAPID; 8 sources in 2 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	reference_stars_catalog_80 (160 sources)		jwst-nirspec-hr	1.5

Reference Stars

Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
1	4	148.944020	69.680289	21.578	1	126	149.008789	69.678671	21.578
1	16	148.941873	69.681669	21.579	1	133	148.944020	69.680289	21.578
1	34	148.957776	69.684452	21.58	1	147	148.941873	69.681669	21.579
1	125	148.992193	69.676730	21.578	1	160	148.994476	69.676761	21.58

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (G395H/F290LP)	c1		148.93049424999998 Degrees 69.67131555555557 Degrees	328.3830222857536			1	1	869.678
2	2 (G235H/F170LP)	c1		148.93049424999998 Degrees 69.67131555555557 Degrees	328.3830222857536			1	1	869.678

Special Requirements	
	Aperture PA Range 325 to 332 Degrees (V3 186.4254303 to 193.4254303) MSA Planned Aperture PA 328.4000 to 328.4000 Degrees (V3 189.8254303 to 189.8254303)